

vSLAM Report-ME 463

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1 Introduction

Simultaneous Localization and Mapping (SLAM) is a core technology in robotics that enables an agent to build a map and localize itself using only onboard sensors [1]. This report presents the results of a monocular vision SLAM implementation based on MATLAB examples [2] and a custom dataset.

2 vSLAM Pipeline

The main steps from vSLAM pipeline are:

- Image sequence download and imageDatastore creation
- Map initialization via two-frame
- Place recognition initialization
- Tracking process
- Local bundle adjustment and map optimization
- Loop closure

2.1 ORB feature matching

ORB feature matching between two frames. Red and green dots represent detected keypoints; yellow lines represent matching points used for pose estimation and 3D point triangulation.

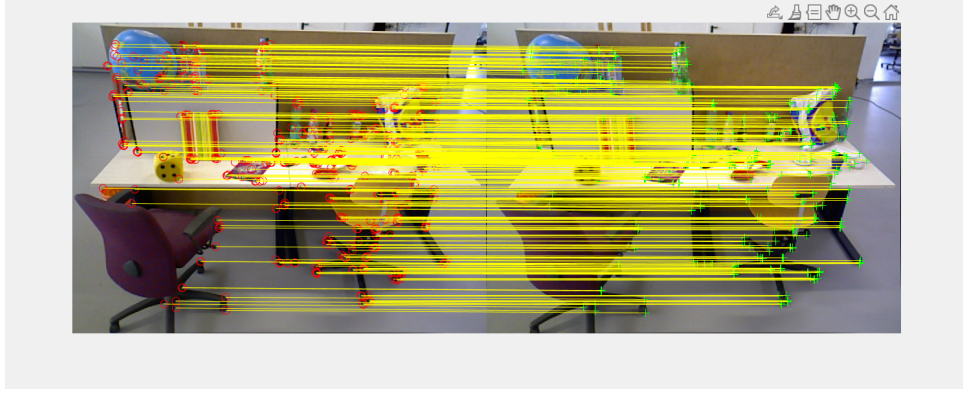


Figure 1: ORB feature matching between two frames

2.2 Place recognition initialization

ORB feature points detected in the initial keyframe. These keypoints are used for matching, triangulation, and pose estimation during SLAM initialization.

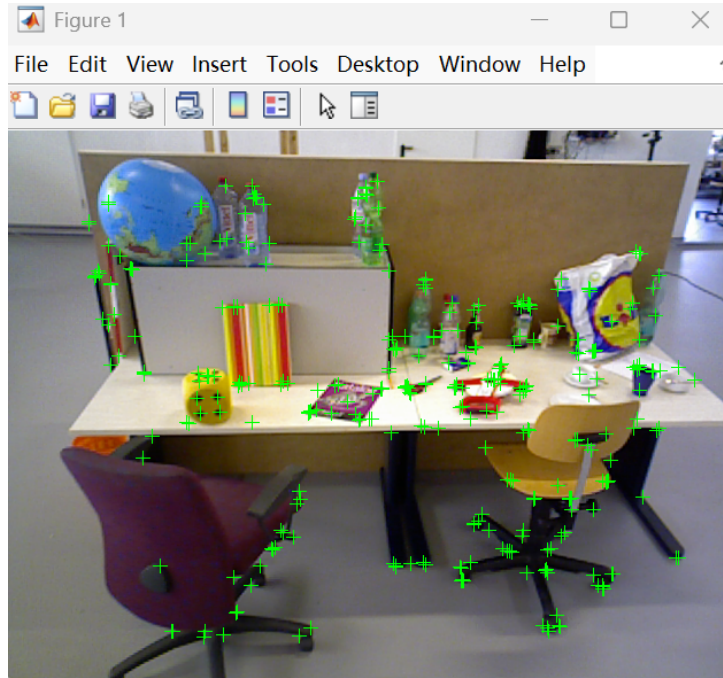


Figure 2: Place recognition

Visualization of the initial map points and camera trajectory after SLAM initialization. The colored points represent the triangulated 3D landmarks and the red camera symbol represents the estimated pose. Keyframe and map point data are managed using *imageviewset* and *worldpointset* objects respectively.

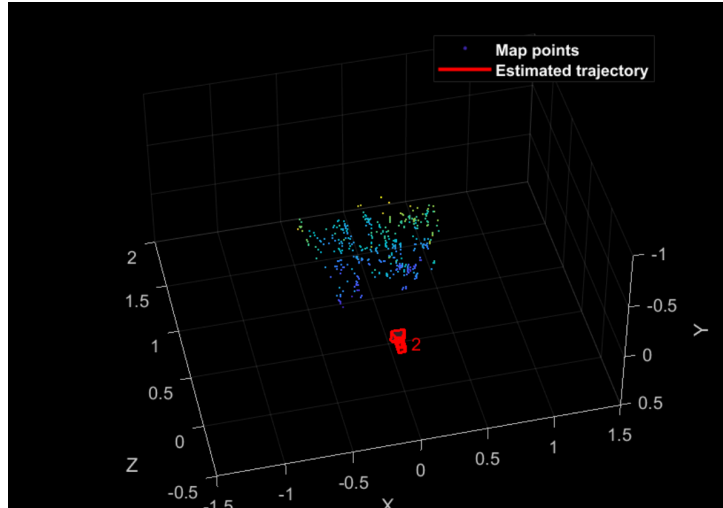


Figure 3: initialization

2.3 Tracking

Use `vision.PointTracker` to track ORB feature points in the current frame. These points are used for camera pose estimation via PnP and further optimized via pure motion bundle adjustment.

After local mapping and loop closure, the 3D map and estimated camera trajectory are reconstructed. Colored points represent 3D landmarks and red curves represent estimated trajectory. Similar pose graph optimization is applied after loop detection to correct drift and update map points.

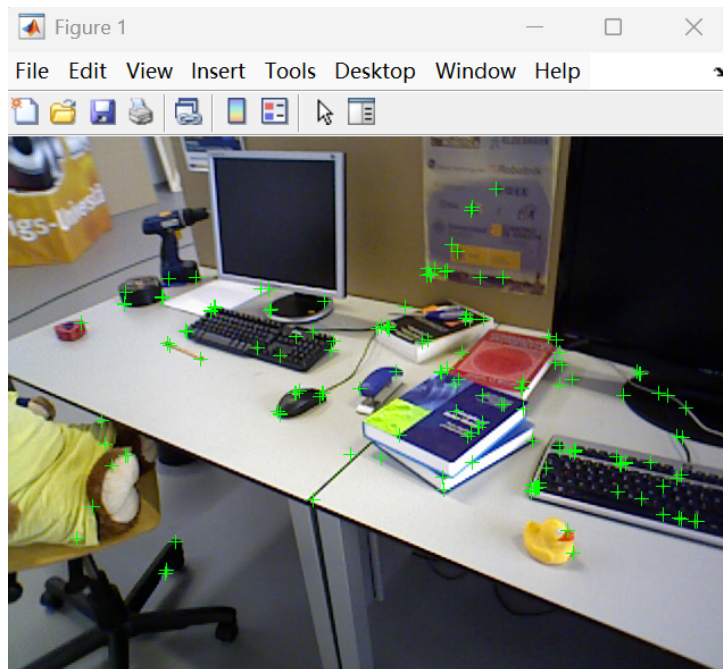


Figure 4: real world

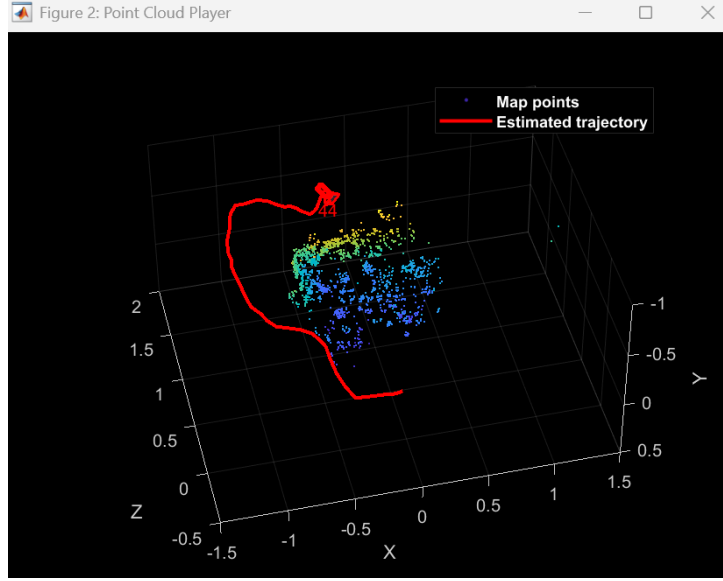


Figure 5: tracking

2.4 Compare with the Ground Truth

Comparison of SLAM trajectories before and after loop closure. The red line shows the estimated trajectory before optimization, the magenta line shows the trajectory after loop closure and similar pose graph optimization, and the green line shows the ground truth. The colored dots represent the reconstructed 3D landmarks. After optimization, the alignment of the trajectory with the ground truth is significantly improved.

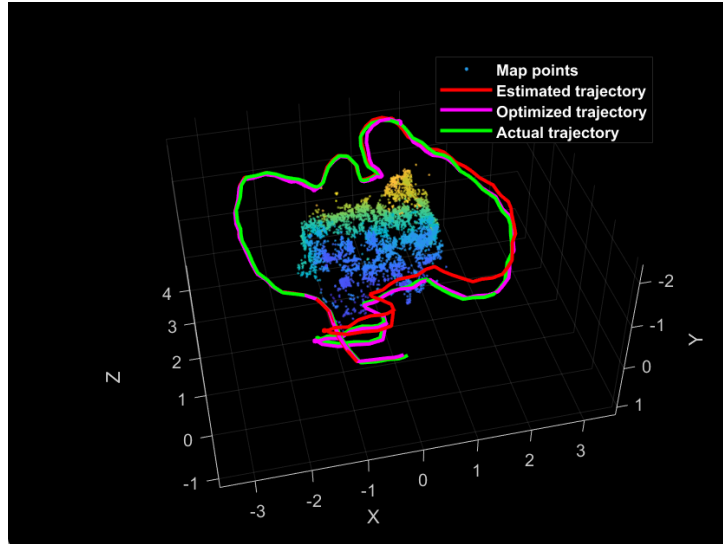


Figure 6: Optimal

3 Part 2: Custom Dataset vSLAM

3.1 Dataset Description

ORB keypoints extracted from the first frame of the custom dataset. The green crosses indicate detected feature points, which are used for inter-frame matching, camera pose estimation, and map initialization. The image was extracted from a video recorded using an iPhone camera.



Figure 7: Recording scene



Figure 8: Place recognition

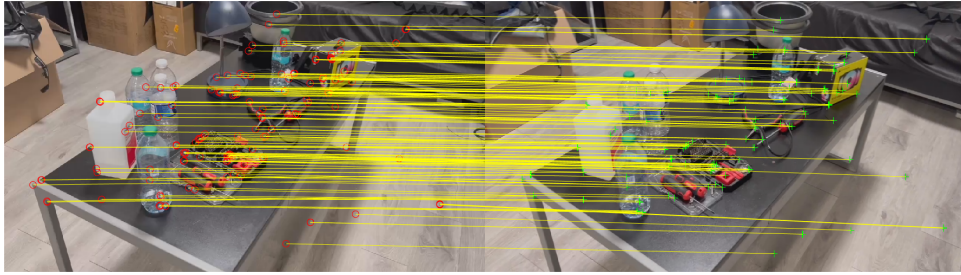


Figure 9: ORB feature matching between two frames

3.2 Camera Calibration

ORB keypoints extracted from the first frame of a custom dataset. Green crosses indicate detected feature points used for image matching, pose estimation, and 3D map initialization. This image was extracted from a video recorded with an iPhone 13 Pro and processed in MATLAB.

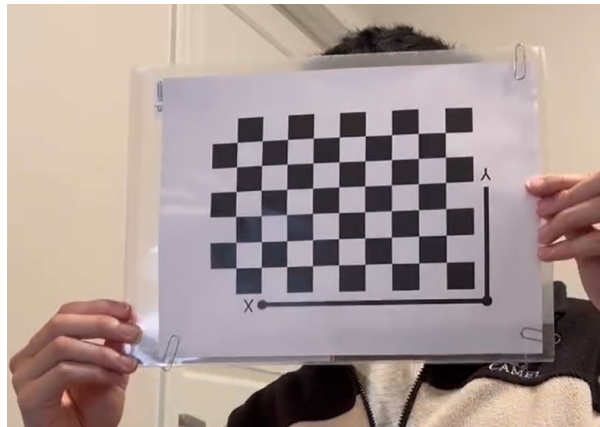


Figure 10: Calibration

```
>> calibrationSession.CameraParameters.K

ans =

1.0e+03 *

    1.0631         0    0.1778
         0    1.0746    0.4778
         0         0    0.0010
```

Figure 11: Parameter of camera

3.3 SLAM Pipeline

Initial SLAM reconstruction after the first two frames. The red curve represents the estimated camera trajectory, and the colored dots represent the triangular 3D landmarks. This initial map can be used as a basis for subsequent tracking and mapping. Estimated

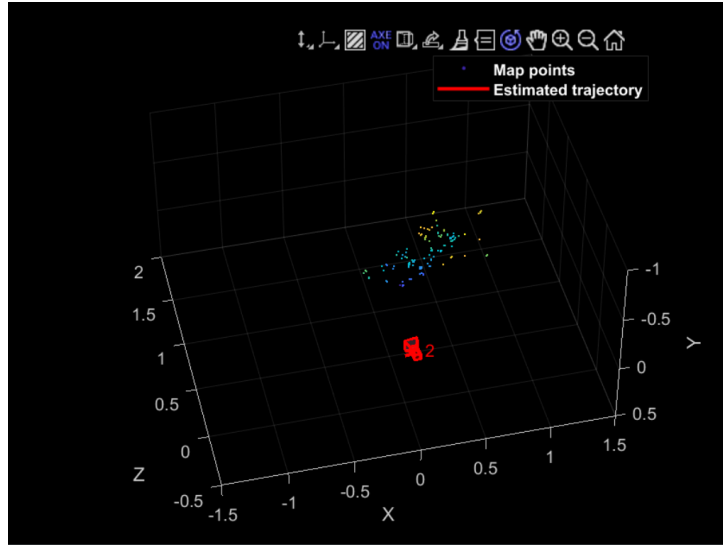


Figure 12: Initialize

trajectory and scatter plot after continuous SLAM tracking. The red trajectory shows the movement of the camera over time, and the dense map point cloud indicates successful tracking.

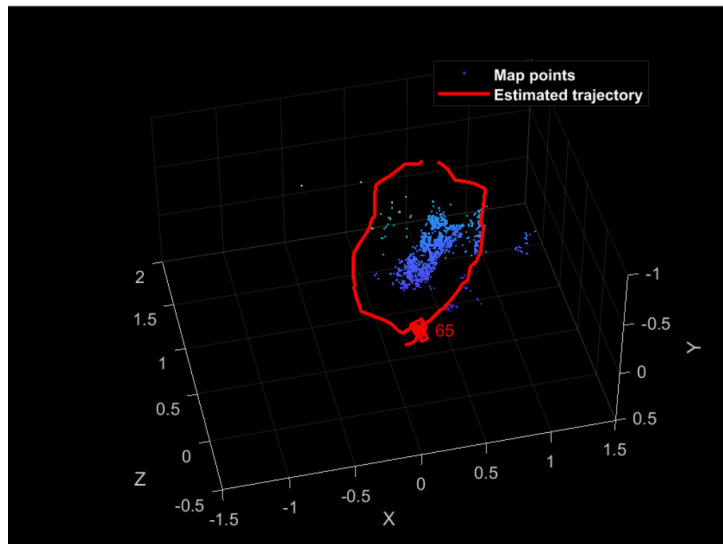


Figure 13: tracking

3.4 Optimal and comparsion

Comparison of SLAM trajectories before and after loop closure and pose graph optimization. The red line represents the unoptimized estimated trajectory, the magenta line represents the trajectory after loop closure. The blue dots represent the reconstructed 3D landmarks. The optimized trajectory is closer to the actual path, thus reducing the accumulated drift.

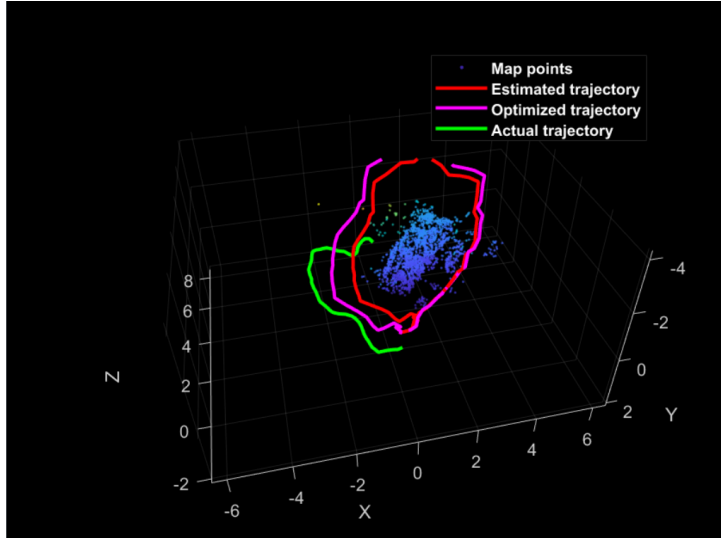


Figure 14: Optimal

4 Conclusion

This project implements a vSLAM system in MATLAB using videos recorded with an iPhone. After fine-tuning frame extraction and adjusting key parameters such as num-SkipFrames, the system successfully completes map initialization and tracking. The results highlight the importance of sufficient features and proper frame sampling for stable SLAM performance.

References

- [1] Hugh Durrant-Whyte and Tim Bailey. Simultaneous localization and mapping: Part i. *IEEE robotics automation magazine*, 13(2):99–110, 2006.
- [2] MathWorks. Visual slam with monocular camera in matlab, 2023.